SolaX Inverter Solution

——External Wind Turbine Solution

ENGLISH

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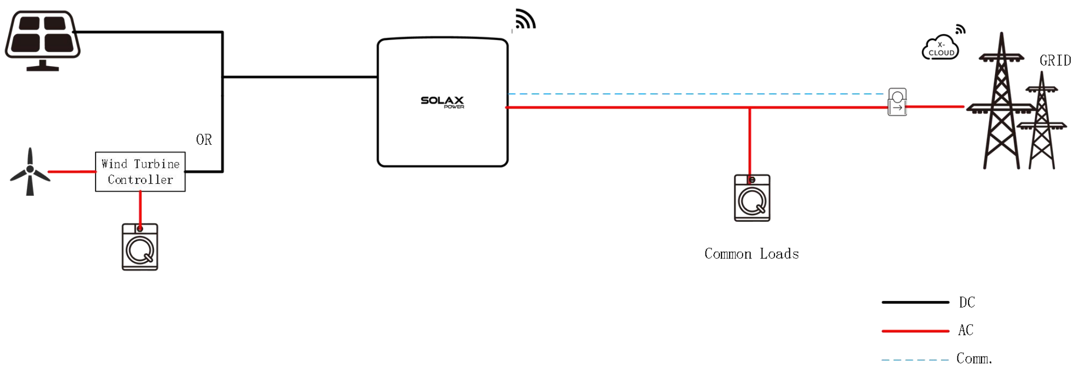
Introduction

With the increasing global demand for renewable energy, wind power, as a clean and efficient form of energy, is becoming an important part of the power system. In order to realize the efficient utilization of wind energy, the combination of energy storage systems and wind turbines is receiving more and more attention.
The scheme of energy storage inverters connected to wind turbines (WT) on the direct current (DC) side can effectively solve the problem of instability and intermittency of wind power generation, and enhance the flexibility and reliability of the power system.
Currently, some regions in Europe, like Poland, are promoting the development of the wind power industry through a variety of policy support measures, including subsidies for wind power inverters, to increase the share of renewable energy in the power system and to achieve the goal of energy transition.
Because of this, the Solax inverter integrates the functions of the connected Wind Turbine, converting wind energy into electricity for use by the load.

Lets take X1-MINI-3.0K-G4 as an example

The following is a diagram of the X1-MINI-G4 external wind turbine solution.



In the **X1-MINI-G4** inverter, PV port (MPPT input) can be connect to WT but only via controller with protective circuits.

NOTE1: Wind turbines are required to have a dedicated controller / self breaker.
Once the:
- load power demand drops
- and grid export is not allowed
- or inverter will have to stop because of grid parameters or any other reason

the speed of the turbine may increase rapidly, which is very dangerous.
The controller of the wind turbine needs to have at least automatic protection, to control produced by WT voltage and power by unloading it if necessary to emergency self breaker (usually resistive load).
That kind of control functions ensures that the wind turbine can be safely connected to the inverter（The third-party controller model we matched tested is

FKJ-GT5KW)

NOTE2: Inverter at PV port can accept only DC parameters so WT AC voltage must first go through appropriate rectifier bridge

NOTE3: When PV port of for example **X1-MINI-G4** is connected to a wind turbine,

- the input DC voltage must not exceed 500 V,
- the current of PV port not exceeding 6A,
- maximum power not exceeding 3kW on PV port,
- minimum starting voltage is 100V

NOTE4: Last point of power curve (VA12) should be set on the maximum level of allowed input power values : 500V and 6A– to be used as a blocking/breaking for WT.

NOTE5: WT controller engaging voltage has to be set 10V above inverter maximum voltage and if controller engaging voltage is smaller than inverter maximum voltage, then inverter maximum blocking voltage and current can be set 10V below

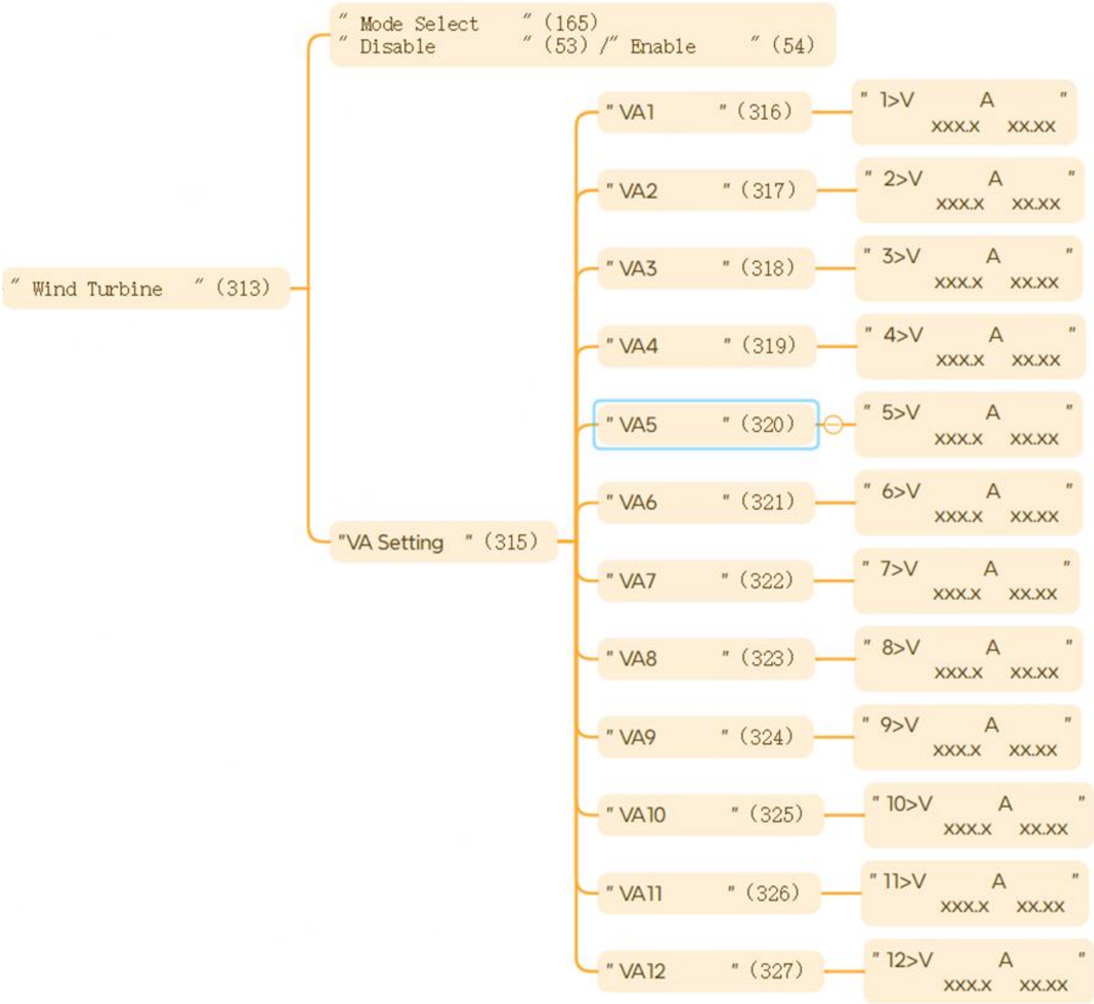
Please note that VA points are different for different WT and must be declared by WT producer.

The table below shows the maximum current depending on the voltage of the wind turbine

Voltage (V)	Max. Current (A)	Max Power (W)
150	6	900
170	6	1020
200	6	1200
250	6	1500
300	6	1800
350	6	2100
400	6	2400
500	6	3000

Setting Wind Turbine

After physically connecting the wind turbine to the inverter via the PV terminals, you need to complete the following setup via the LCD.
NOTE: for setting



a. Select Settings > Wind Turbine, and set Wind Turbine to Enabled.

b. Select VA Setup, where you can choose Wind Turbine to set the VA1~VA12 values individually.

When setting values for each VA, with X1-MINI-G4 you must note that:
» The voltage must increases progressively. That is, V1<V2<V3 … <V12. The voltage setting range is between 0 and 500 V.
» Current values A1~A12 can be set irregularly:
Current value for PV must be less than 6 A.

Note: Adjust the settings according to the actual model and wind turbine performance.

The Solax Cloud and app also support the setup of（coming soon）。

Supported Inverter Models and Software Version

Model	Progress	Software Version
X1-MINI-G4	Done	arm ≥ V35， dsp ≥V28

Special Note

» The voltage must increases progressively. That is, V1<V2<V3 … <V12. The voltage setting range is between 0 and 500 V.

» Current values A1~A12 can be set irregularly:

Current value for PV port must be less than 6 A.

Operation Notes

1. Before starting the installation, make sure that the Wind Turbine is in a power-off state before connecting it;
2. Make sure that the grounding of the Wind Turbine controller and inverter is completely reliable.